

# Power Plant Engineering (ME 4201)

## Design of a 20 kW Grid-Connected SPV-Wind Hybrid Power Plant in Char Kukri Mukri



**Submitted by:**

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## Work Distribution:

| <b>Name</b>                   | <b>ID</b>          | <b>Chapters</b> |
|-------------------------------|--------------------|-----------------|
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## **1.1 Background of the Project**

In the southernmost part of Bangladesh, Char Kukri Mukri is an island in the Bhola district and this island is by the Gangetic delta. Here, the climate is humid with a tropical monsoon. The region named Uttor Kukri Mukri that we selected, benefits from a high solar exposure which makes this place perfect for photovoltaic applications along with having moderate wind speed to harness wind energy generation throughout the year.[1]

Solar and wind energy together provides a great power supply. Solar energy is generally acute in the daytime and wind speed is variable depending on seasons and in night time, it is generally higher. National grid codes and standards should be strictly maintained for the grid connectivity. By implementing a 20 kW grid-connected solar PV- wind power plant in Char Kukri Mukri can increase energy access, makes the environment sustainable and also helps to develop the social and economic arena of our country. The challenges that come with it, can be handled by correct planning and involvement with the community for ensuring the project's success.

## **1.2 Type of Project**

If a 20kW grid connected SPV wind hybrid powerplant can be successfully designed and built, then it will highly increase the energy supply for the prime consumers like residential areas, schools, micro enterprises and mosques.

Solar energy stays at the peak during daytime which perfectly aligns with the ongoing school timing and the running hours of the micro enterprises. As for wind energy, it can be harnessed both in day and night time. This makes wind energy provide a consistent power supply. So, rather than depending on a single power source, reliability on both solar and wind power source enhances the electricity generation to greater heights and thus helps it to be efficient and sustainable for the environment. It is also mentionable that the power supply and consumption timing readily aligns with each other and thus it can meet the present and future electricity demands of Char Kukri Mukri by making it a sustainable energy infrastructure.

## **1.3 Location**

The exact coordinates of the location where we want to establish the 20 kW grid-connected solar photovoltaic (SPV) and wind hybrid power plant near Uttar Char Kukri Mukri Government Primary School is 21°57'02.8"N and 90°39'14.8"E.[2]

The need for a reliable and sustainable energy source would be sufficed by establishing this power plant and it would also help to lessen the dependency on the national grid which would help to lower the electricity bills for the residents and also educational institutions, micro-enterprises, and places of worship. If we can harness these renewable energy efficiently for the lights, fans, and other appliances, we can increase the sustainability of our environment by reducing greenhouse gas emissions significantly.



Figure 1.1: A Google maps view of the selected location for the plant

## Chapter 2

### Demand Analysis

#### 2.1 Load Demand

Load demand refers to the total amount of electrical power required by consumers at any given time. It fluctuates based on factors like time of day, season, weather conditions, and the type of consumers (residential, industrial, commercial). Helps power plants generate only the required electricity, avoiding waste. Ensures proper load distribution among power stations to prevent overloading. The data provided below comprises the power consumption of different electrical components and their total energy usage over a day, measured in watt- hours.

Table 2.1 demonstrates the components and their operating hours and total consumption during summer.[4]

## Summer

| Component           | Quantity | Power (W) | Operating Hours | Total Consumption<br>(Wh/day) |
|---------------------|----------|-----------|-----------------|-------------------------------|
| Lighting (30 W)     | 18       | 30        | 6               | 3240                          |
| Fan (70 W)          | 6        | 70        | 12              | 5040                          |
| Computer (200 W)    | 6        | 200       | 4               | 4800                          |
| Refrigeration(100W) | 2        | 100       | 24              | 4800                          |
| Air Conditioner(    | 2        | 120       | 6               | 1440                          |
| Other Appliances    |          | 500       | 2               | 1000                          |

The components listed above in the table reflects the energy consumption in the residential area each with separate power ratings in the summer season :

### 1. Lighting:

In the given table, 18 bulbs of 30W are operated for 6 hours per day and thus results in total energy consumption of 3240 Wh/ day.

### 2. Fan:

6 fans of 200 W are operated for 4 hours per day and the total energy consumption is 4800 Wh/ day

### 3. Computer:

6 computers of 200 W are operated for 4 hours daily and the total energy consumption is 4800 W.

**4. Refrigerator:**

2 refrigerators of 100W are operated for 24 hours daily and the total energy consumption is 4800 W.

**5. Air Conditioner:**

2 air conditioners of 120 W are operated for 6 hours daily and the total energy consumption is 1440 W.

**6. Other Appliances:**

Some other household appliances of 500W are operated for 2 hours daily and the total energy consumption is 1000 W.

Table 2.2 demonstrates the components and their operating hours and total consumption during winter.

## Winter

| Component           | Quantity | Power (W) | Operating Hours | Total Consumption (Wh/day) |
|---------------------|----------|-----------|-----------------|----------------------------|
| Lighting (30 W)     | 18       | 30        | 6               | 3240                       |
| Fan (70 W)          | 6        | 70        | 2               | 840                        |
| Computer (200 W)    | 6        | 200       | 4               | 4800                       |
| Refrigeration(100W) | 2        | 100       | 24              | 4800                       |
| Air Conditioner(    | 2        | 120       | 0               | 0                          |

|                  |  |     |   |      |
|------------------|--|-----|---|------|
| Other Appliances |  | 500 | 2 | 1000 |
|------------------|--|-----|---|------|

The components listed above in the table reflects the energy consumption in the residential area each with separate power ratings in the winter season :

**1. Lighting:**

In the given table, 18 bulbs of 30W are operated for 6 hours per day and thus results in total energy consumption of 3240 Wh/ day.

**2. Fan:**

In the given table, 6 fans of 70W are operated for 2 hours per day and thus results in a total energy consumption of 840 Wh/ day.

**3. Computer:**

In the given table, 6 computers of 200W are operated for 4 hours per day and thus results in a total energy consumption of 4800 Wh/ day.

**4. Refrigerator:**

In the given table, 2 refrigerators of 100W are operated for 24 hours per day and thus results in total energy consumption of 4800 Wh/ day.

**5. Air Conditioner:**

In the given table, 2 air conditioners of 120W are not operated at all in the whole 24 hours.

**6. Other Appliances:**

In the given table, some household appliances of 50W are operated for 2 hours per day and thus results in a total energy consumption of 1000 Wh/ day.

## 2.2 Daily Load Curve

The daily load curve gives information about the load on the power station during different running hours of the day and the area under the daily load curve gives the total units of electrical energy generated. During the summer season at the 18th hour, peak energy consumption is 18800 W and during the winter season at the same hour, peak energy consumption is 1400 W.

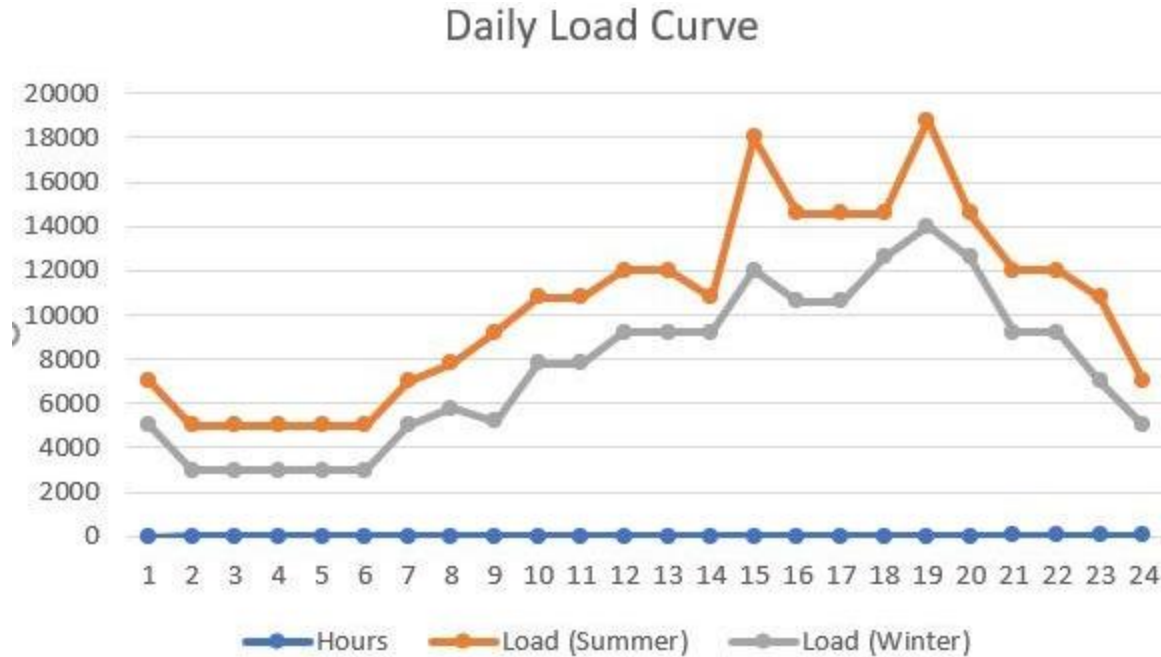


Figure 2.1: Daily load curve

## Chapter 3 Resource Assessment

### 3.1 Detailed Resources Assessment

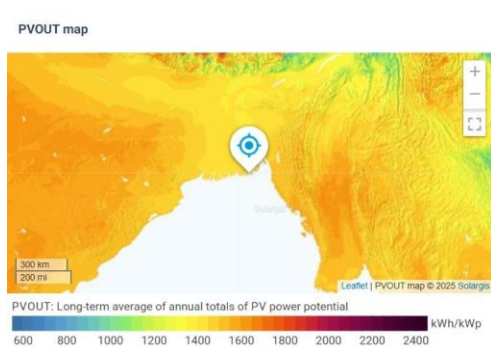
As the location for the SPV wind hybrid power plant has been finalised, the sustainability for meeting the necessary energy demand has to be analysed now. Data from the Global Solar Atlas website has been put into action for making an initial assessment of the SPV electricity for this specific area in Char Kukri Mukri.[4]

#### 3.1.1 Site Information

**Project location:** Char Kukri Mukri, Bhola, Bangladesh  
**Geographical Coordinates:** 21°57'02.8"N, 90°39'14.8"E  
**Time zone:** UTC+6, Asia/ Dhaka [BST]  
**Elevation:** 1m ( 6 feet above sea level)



**Fig 3.1 : Wind Resource Map of the location [4]**



**Fig 3.2:** Satellite map and PV Output Map of the location [3]

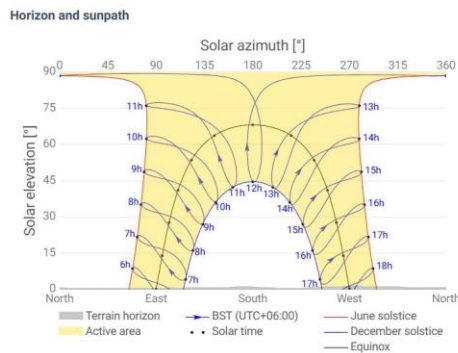
### 3.1.2 Site info for Solar PV

The below chart demonstrates the map data from Global Solar Atlas for the site in Char Kukri Mukri.

| Map data                                                                                                             |                | Per year                  |
|----------------------------------------------------------------------------------------------------------------------|----------------|---------------------------|
|  Specific photovoltaic power output | PVOUT specific | 1484.9 kWh/kWp            |
| Direct normal irradiation                                                                                            | DNI            | 1175.7 kWh/m <sup>2</sup> |
| Global horizontal irradiation                                                                                        | GHI            | 1750.0 kWh/m <sup>2</sup> |
| Diffuse horizontal irradiation                                                                                       | DIF            | 918.8 kWh/m <sup>2</sup>  |
| Global tilted irradiation at optimum angle                                                                           | GHI opta       | 1868.4 kWh/m <sup>2</sup> |
| Optimum tilt of PV modules                                                                                           | OPTA           | 23 / 180                  |
| Air temperature                                                                                                      | TEMP           | 26.2 °C                   |
| Terrain elevation                                                                                                    | ELE            | 3 m                       |

**Fig 3.3** : Map data of Char Kukri Mukri, Bhola (per year) [4]

The below figure demonstrates the sun path diagram for the site in Char Kukri Mukri.

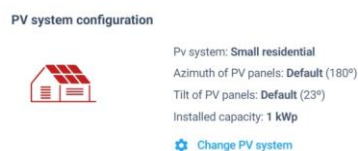


**Fig 3.4** : Sun Path and Azimuth Diagram of the site [4]

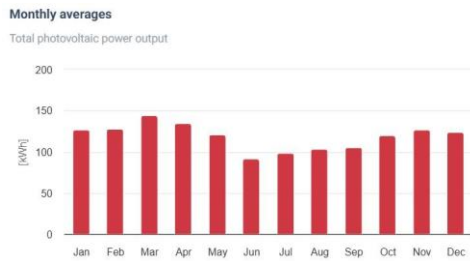
This sun path diagram shown in figure 3.4 is completed by plotting solar azimuth and elevation angles of a typical meteorological year for every hour of the day. It is seen that in June, maximum solar elevation ( 90° ) is found at noon and the azimuth direction is north. But in December noontime solar elevation angle is ( 45° ).

### 3.1.3 PV Electricity and Solar Radiation

The PV system configuration used to acquire data from Global Solar Atlas has been given below.



**Fig 3.5 to Fig 3.8** shows the photovoltaic power output and DNI trends of the location from the Global Solar Atlas report.

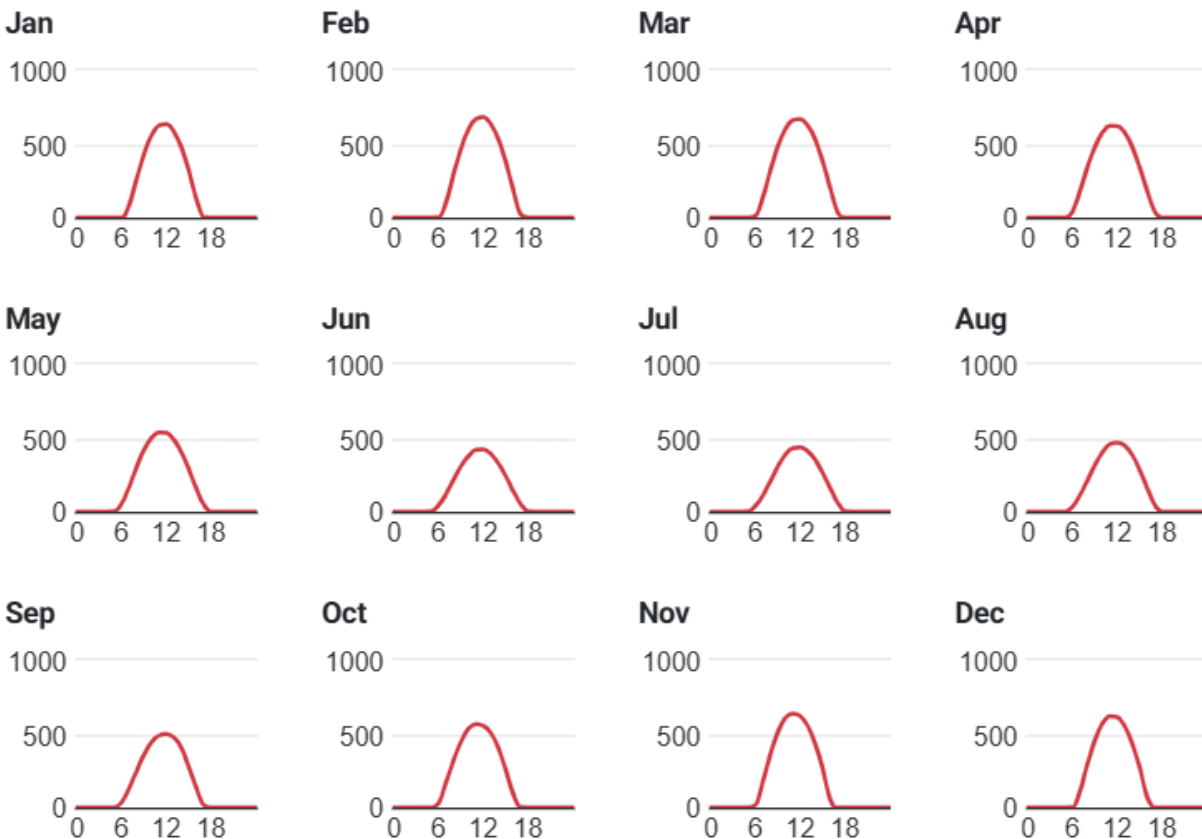


**Fig 3.6:** Monthly averages of total photovoltaic power output [4]

Fig 3.6 depicts the amount of energy generated every month from a 10kW system. In March, the maximum power output was found to be about 140 kWh. In monsoon months, the power output is the lowest in June.

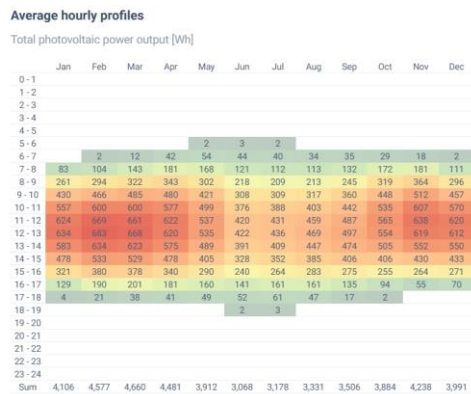
### Average hourly profiles

Total photovoltaic power output [Wh]



**Fig 3.7 :** Average hourly profile of total photovoltaic power output for every month in Wh.[4]

Here, the PV power output of every month is plotted against the hours of the day and from this curve, the peak power hours for every day throughout the year is noon.



**Fig 3.8 :** Average hourly profile of total photovoltaic power output in Wh.[4]

In this above chart, the number of energy units produced per hour when the system operates at full capacity is depicted. For instance, in April, from 6 a.m. to 7 a.m. , 42 Wh of energy is produced and the highest amount of energy is 622 Wh during 11 a.m. to 12 p.m.

### 3.1.4 Area Data for Wind

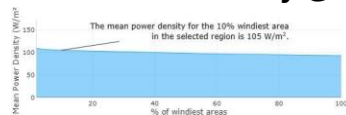
From figure 3.9 to figure 3.14 shows the acquired data from Global Wind Atlas for center (Lat, Long): 21°57'02.8", 90°39'14.8" at Uttor Kukri Mukri. [6]



**Fig 3.9:** Mean power output and wind speed [5]

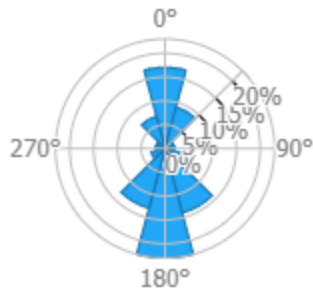
Data from the Global Wind Atlas states that from the selected location the mean power output is 105 W/m<sup>2</sup> and the mean wind speed is 4.38 m/s for 10% of windiest areas at 50m height.

### Mean Power Density @Height 50m



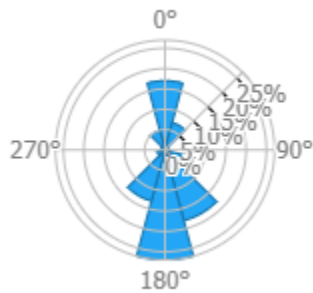
**Fig 3.10:** Power density [5]

Fig 3.10 shows the mean power density for a particular percentage range. For example, the 10% windiest area in the selected regions' mean power density is 105 W/m<sup>2</sup>.



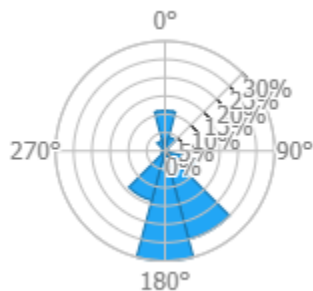
**Fig 3.11: Wind Frequency rose [5]**

It shows the frequency of winds blowing from the south direction throughout the year.



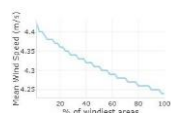
**Fig 3.12: Wind Speed Rose [5]**

This figure shows that the general wind direction and maximum wind speed are in the south or southwest direction throughout the year.



**Fig 3.13: Wind Power rose [5]**

This figure shows the wind power direction throughout the year in this case, maximum wind power is found from the south direction.

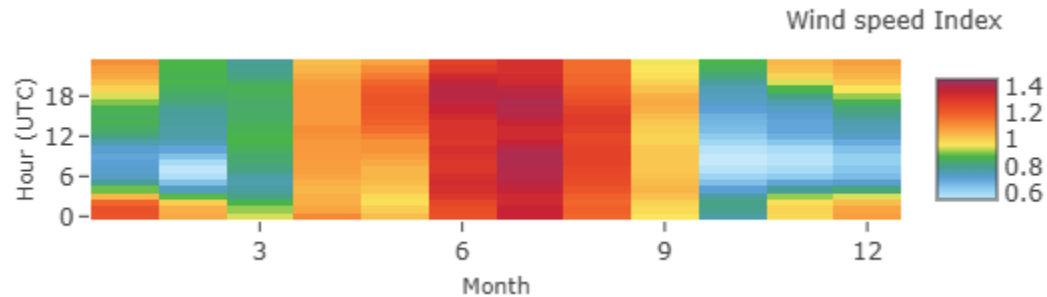


**Fig 3.14: Wind Speed [5]**

This figure shows the mean wind speed vs the percentage of windiest areas. For example, 10% windiest area in the selected regions means wind speed is around 4.38 m/s.

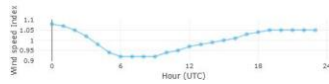
### 3.1.4 Temporal Data for Wind

Figure 3.15 to figure 3.19 depicts the temporal data acquired from Global Wind Atlas for the specified location of the plant.



**Figure 3.15:** Heatmap [5]

This figure shows the speed of wind found throughout the whole year. For example, in July from 6 am to 12 pm velocity of wind will be very high which is around 1.4, but in October at the same time range, the velocity will be low (around 0.6).



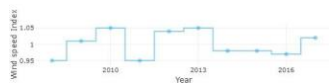
**Figure 3.16:** Hourly Bar Chart [5]

Figure 3.16 shows the wind speed index of every month plotted against the hours of the day. The wind index indicates how fast the average wind speed is over a certain period in comparison to the typical one. Around 100% wind index depicts the average wind speed during the tested period was higher or lower than normal. From this curve, the peak speed hour every day in the full year is evening.



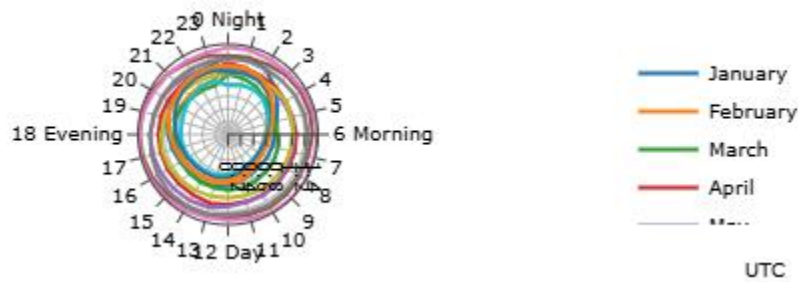
**Figure 3.17:** Hourly Bar Chart [5]

Figure 3.17 shows the wind speed index vs the month graph. For example, the wind velocity will be highest in July and in October, it will be the lowest.



**Figure 3.18:** Annual bar chart [5]

This figure shows the speed of wind every year from 2008 to 2017. For instance, in 2010 and 2013, the average wind speed is 1.05 m/s and in 2016, the wind speed is around 0.97 m/s.



**Figure 3.19:** Hourly vs Monthly radar plot [5]

This figure is used to compare a group of variables to another group of the same variables. Like here we compare the hourly wind speed with the monthly wind speed.

### 3.2 Key Findings

The main findings from the preliminary resource assessment of Uttor Char Kukri Mukri Bhola , Bangladesh are listed below:

- The optimum tilt angle is  $23^\circ$  and the optimum azimuth angle is  $180^\circ$  (figure 3.3)
- The total annual PV power output is 1.426 MWh/ year and the annual Global tilted irradiation is 1868.4 kWh/m<sup>2</sup>.
- The annual average DNI value is 1175.7 kWh/m<sup>2</sup>. ( figure 3.3)
- The range of mean power density is around 105 W/m<sup>2</sup>. (Figure 3.10)
- The optimum direction to place a wind turbine is south. (Figure 3.14)
- From May through August, the most power will be generated. However, electricity generation will be low in October, November, December, and February. (Figure 3.17)
- In the chosen location for the 10% windiest areas at 50m height, the mean power output is near 105W/m<sup>2</sup> and the mean wind speed is 4.38 m/s. (Figure 3.9)

## Chapter 4

### Selection of Conversion Technology

#### 4.1 Overall System

The overall system is a 20KW solar PV-Wind Hybrid Power Plant. The system was designed in such a way that it can be suitable for the selected location in Char Kukri Mukri [1]. As the powerplant is designed as a hybrid system, it ensures an uninterrupted power supply, low fuel consumption, etc benefits [12]. Thus, the selected location can be enriched like never before for the irrigation sector. We decided to follow the design based on a paper named “Optimized Design of a Hybrid PV-Wind-Diesel Energy System for Sustainable Development at Coastal Areas in Bangladesh” [12]. The diagram based on the paper is given below:

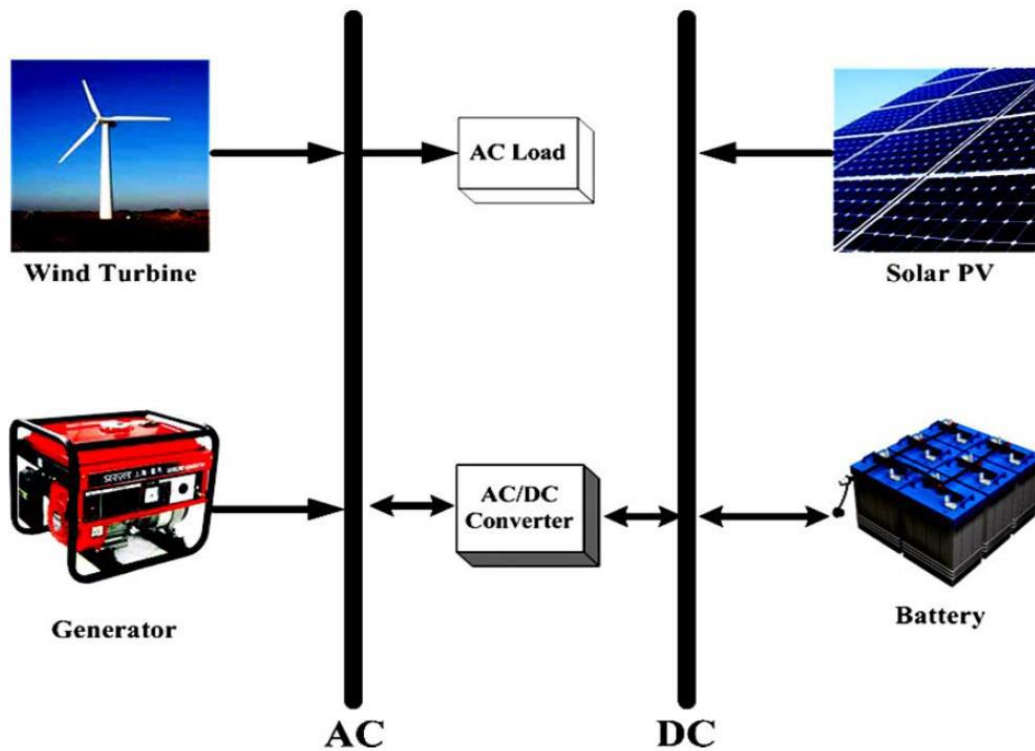


Figure 4.1: Schematic block diagram of a hybrid solar PV-wind energy system.

#### 4.2 Selection of Components through Literature Review

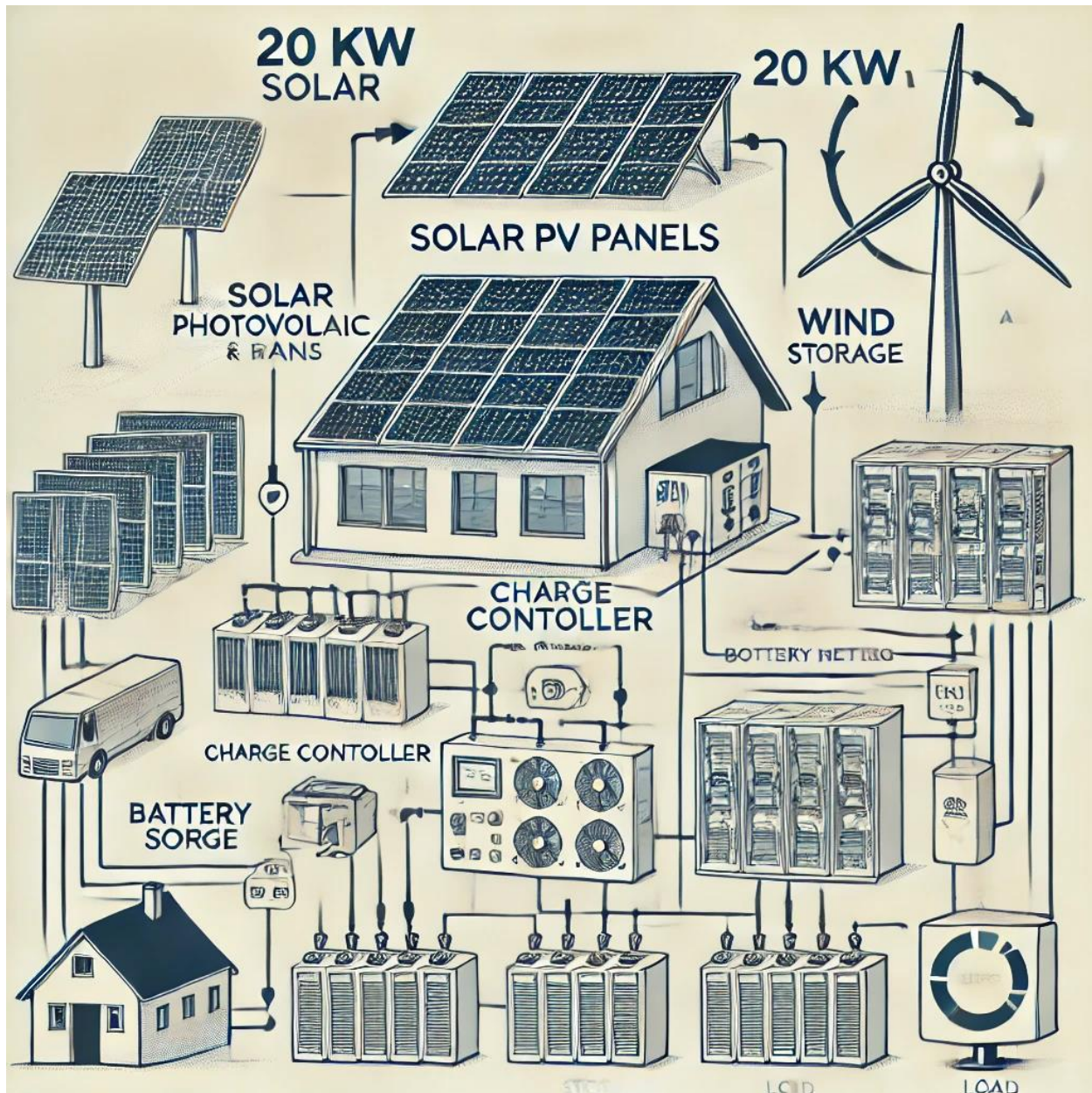
| SI NO. | Components        | No. of Units | Price/Unit(BDT) | Total Price |
|--------|-------------------|--------------|-----------------|-------------|
| 1      | Solar PV          | 70           | 17842           | 1248940     |
| 2      | Wind Turbine      | 1            | 2500000         | 2500000     |
| 3      | Charge controller | 30           | 90000           | 2700000     |
| 4      | Inverter          | 2            | 144000          | 288000      |
| 5      | Battery Charger   | 4            | 22400           | 89600       |
| 6      | Battery           | 10           | 40000           | 400000      |
| 7      | Diesel Generator  | 1            | 450000          | 450000      |

The components have been selected based on the paper and the research on the internet. The prices have also been selected by following these references [11]- [13 ]. These components are of immaculate importance to build a hybrid powerplant for the long run and even though the initial cost may be high, the benefits are also on par.

## Design & Optimization

The Figure below shows the preliminary designs for a SPV-Wind plant which will be designed.

## 5.1 Preliminary Design



**Figure 5.1: Basic Outline of a PV-Wind Hybrid Plant [11]**

### 5.2.1 Performance Analysis of 10kW SPV system

A hybrid powerplant plant with a 20kW capacity can produce the following amounts of energy using the following angles. Here, the slope angle is not optimized and is considered 23° as per the latitude of the location. The system loss was considered to be 14%, adjusting to real-life conditions.

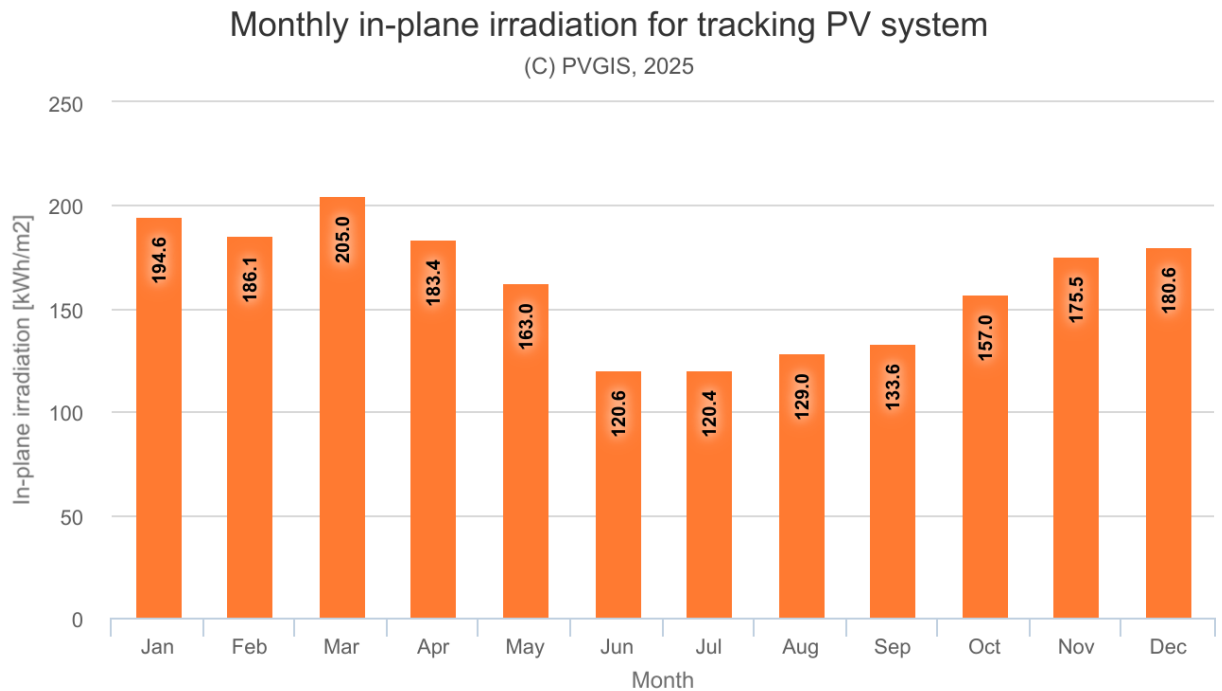
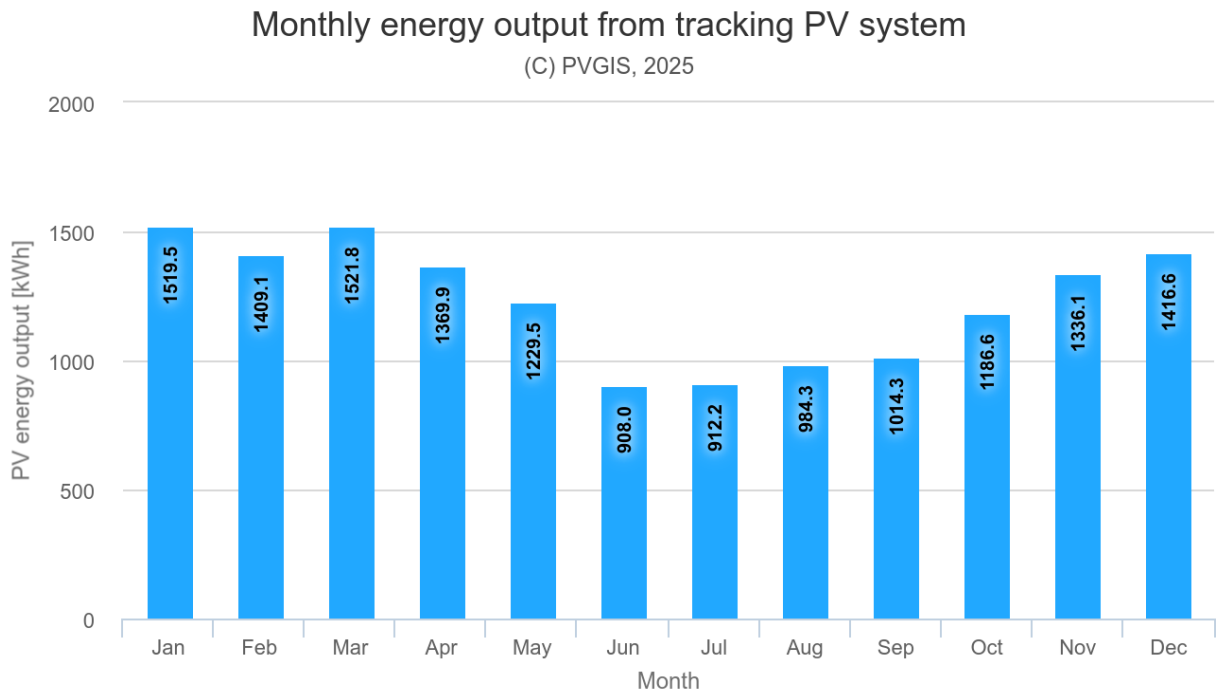
| Provided inputs :   |                     |
|---------------------|---------------------|
| Location [Lat/Lon]: | 21.948 , 90.650     |
| Horizon:            | Calculated          |
| Database used:      | PVGIS-ERA5          |
| PV technology:      | Crystalline silicon |
| PV installed [Wp]:  | 10                  |
| System loss [%]:    | 14                  |

| Simulation outputs :                               |          |
|----------------------------------------------------|----------|
| Slope angle [°]:                                   | 23       |
| Azimuth angle [°]:                                 | 0        |
| Yearly PV energy production [kWh]:                 | 14807.79 |
| Yearly in-plane irradiation [kWh/m <sup>2</sup> ]: | 1948.68  |
| Year-to-year variability [kWh]:                    | 298.09   |
| Changes in output due to:                          |          |
| Angle of incidence [%]:                            | -2.58    |
| Spectral effects [%]:                              | 0.48     |
| Temperature and low irradiance [%]:                | -9.74    |
| Total loss [%]:                                    | -24.01   |

**Figure 5.2: Simulation Data from PVGIS without Optimization [13]**

The following bar chart extracted from the PVGIS report shows the distribution of solar energy output for a year. March has the maximum output using the corresponding angle mentioned above.



**Figure 5.3: Solar Energy Output throughout the year. [13]**

### 5.2.2 Performance Analysis of 10kW HAWT system

Using resource assessment data from chapter 3, the monthly mean wind speed was given as input in RETScreen Expert as seen in figure 5.8. The monthly wind speed is calculated using the wind speed index (figure 3.17) and the mean wind speed of 4.38m/s (figure 3.9).

Wind - Level 3

**Resource assessment**  
Resource method

Wind speed

Atmospheric pressure Air temperature

Climate Data  
Canada - Quebec - Varennes

Electricity rate – annual  
\$/kWh

Electricity production  
kWh

| Month         | Wind speed<br>m/s | Atmospheric pressure<br>kPa | Air temperature<br>°C | m/s        | kPa          | °C         | \$/kWh      | kWh           |
|---------------|-------------------|-----------------------------|-----------------------|------------|--------------|------------|-------------|---------------|
| January       | 5.3               | 100.9                       | -9.6                  | 4.7        | 100.9        | -9.6       | 0.09        | 993.985       |
| February      | 4.5               | 100.9                       | -7.8                  | 4.5        | 100.9        | -7.8       | 0.09        | 558.355       |
| March         | 4.8               | 100.9                       | -2.3                  | 4.6        | 100.9        | -2.3       | 0.09        | 748.489       |
| April         | 6.2               | 100.7                       | 5.6                   | 4.4        | 100.7        | 5.6        | 0.09        | 1,373.130     |
| May           | 6.3               | 100.7                       | 13.4                  | 4.2        | 100.7        | 13.4       | 0.09        | 1,431.649     |
| June          | 7.6               | 100.5                       | 18.9                  | 3.6        | 100.5        | 18.9       | 0.09        | 1,944.012     |
| July          | 7.8               | 100.6                       | 20.5                  | 3.2        | 100.6        | 20.5       | 0.09        | 2,109.617     |
| August        | 7.0               | 100.7                       | 19.5                  | 2.8        | 100.7        | 19.5       | 0.09        | 1,745.659     |
| September     | 5.8               | 100.9                       | 15.1                  | 3.1        | 100.9        | 15.1       | 0.09        | 1,131.674     |
| October       | 4.0               | 100.9                       | 8.3                   | 3.7        | 100.9        | 8.3        | 0.09        | 410.985       |
| November      | 4.5               | 100.9                       | 2.1                   | 4.2        | 100.9        | 2.1        | 0.09        | 576.938       |
| December      | 4.9               | 100.9                       | -4.5                  | 4.4        | 100.9        | -4.5       | 0.09        | 802.595       |
| <b>Annual</b> | <b>5.7</b>        | <b>100.8</b>                | <b>6.7</b>            | <b>3.9</b> | <b>100.8</b> | <b>6.7</b> | <b>0.09</b> | <b>13,827</b> |

Measured at  
m

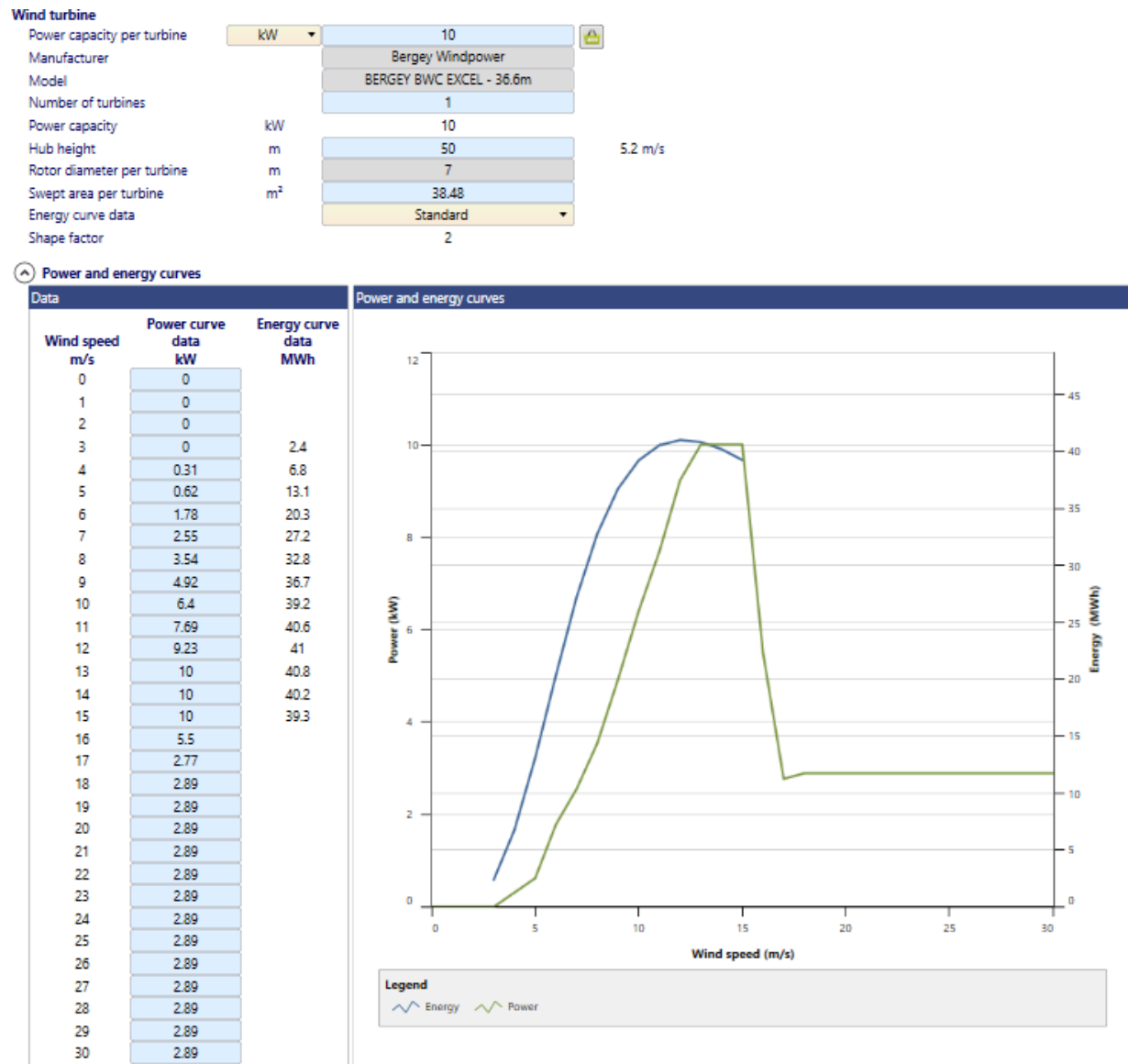
Wind shear exponent  
100

0.14

**Figure 5.4: Resource assessment of a 10kW wind turbine for Char Kukri Mukri**

RETScreen Expert uses its resources to estimate the production of electricity. For the selected location energy exported to the grid would be 13827 kWh annually.

Figure 5.9 shows the power output and energy output curve of Bergey Windpower's Excel-10 model with a hub height of 50m.



**Figure 5.5: Power and Energy curve of the 10kW Wind turbine in Char Kukri Mukri**

From figure 5.9, for a standard energy curve, Rayleigh Distribution is considered. For a maximum of 10kW power, cut-off wind speed is found around 16m/s. The simulation results for the provided inputs are shown in figure 5.10.

### Losses

|                      |   |     |
|----------------------|---|-----|
| Array losses         | % | 2%  |
| Airfoil losses       | % | 3%  |
| Miscellaneous losses | % | 1%  |
| Availability         | % | 98% |

### Summary

|                     |            |        |
|---------------------|------------|--------|
| Capacity factor     | %          | 15.8%  |
| Initial costs       | \$/kW      | 6,400  |
|                     | \$         | 64,000 |
| O&M costs (savings) | \$/kW-year | 48     |
|                     | \$         | 480    |
| Energy saved        | kWh        | 13,827 |

### Other information

|                              |                    | Per turbine |
|------------------------------|--------------------|-------------|
| Unadjusted energy production | kWh                | 14,836      |
| Pressure coefficient         |                    | 0.995       |
| Temperature coefficient      |                    | 1.030       |
| Gross energy production      | kWh                | 14,992      |
| Losses coefficient           |                    | 0.92        |
| Specific yield               | kWh/m <sup>2</sup> | 359         |

**Figure 5.6: Stimulation results for the provided inputs**

Figure 5.10 shows the capacity factor for 50m 10kW HAWT using the Bergey Excel-10 model would be 15.8% and annual energy production is 14,836 kWh.

### 5.3.1 Optimization Process for the SPV system

After analyzing the performance of the 10kW ground-mounted solar PV plant, further analysis was done to optimize its performance. Optimization of the azimuth angle and tilt angle were done using the PVGIS website [46] in an attempt to further enhance the output results.

| Provided inputs :                                  |                     |
|----------------------------------------------------|---------------------|
| Location [Lat/Lon]:                                | 21.948 , 90.650     |
| Horizon:                                           | Calculated          |
| Database used:                                     | PVGIS-ERA5          |
| PV technology:                                     | Crystalline silicon |
| PV installed [Wp]:                                 | 10                  |
| System loss [%]:                                   | 14                  |
| Simulation outputs :                               |                     |
| Slope angle [°]:                                   | 27                  |
| Azimuth angle [°]:                                 | 2                   |
| Yearly PV energy production [kWh]:                 | 14843.43            |
| Yearly in-plane irradiation [kWh/m <sup>2</sup> ]: | 1954.1              |
| Year-to-year variability [kWh]:                    | 296.11              |
| Changes in output due to:                          |                     |
| Angle of incidence [%]:                            | -2.56               |
| Spectral effects [%]:                              | 0.48                |
| Temperature and low irradiance [%]:                | -9.79               |
| Total loss [%]:                                    | -24.04              |

**Figure 5.7: Provided inputs for optimization and simulation outputs from PVGIS**

To optimize the slope angle and azimuth angle, system inputs were given in the grid-connected PV tab, as the optimization option was not available for off-grid PV systems. A system loss of 14% was considered as it was the default value. The simulated results show that the optimum tilt angles of the panels will be 27° and the optimum azimuth angle is 2° (south to west).

## 5.4 Technical Specifications of Final Design

To provide the model names, manufacturing details, and additional small details for each component, I will create an updated table with more specific information. The details will include a general example of model names, manufacturers, and some additional technical aspects. However, please note that specific model names and manufacturers may vary depending on the actual hardware used.

Here's an example of what the technical details might look like for each component:

### 1. Solar PV Module [17]

- **Model Name:** SunPower SPR-X22-370
- **Manufacturer:** SunPower Corporation
- **Power Output:** 370W
- **Efficiency:** 22.7%
- **Panel Type:** Monocrystalline
- **Warranty:** 25 years
- **Additional Info:** High efficiency, durable performance in diverse environmental conditions.

### 2. Wind Turbine [17]

- **Model Name:** BERGEY BWC EXCEL - 36.6m
- **Manufacturer:** Bergey Windpower
- **Rated Power Output:** 8.9 kW
- **Rotor Diameter:** 7 meters
- **Cut-in Wind Speed:** 2.5 m/s
- **Warranty:** 5-10 years
- **Additional Info:** High reliability and designed for both onshore and offshore applications.

### 3. Diesel Generator [17]

- **Model Name:** Cummins C30D5
- **Manufacturer:** Cummins Inc.
- **Rated Power Output:** 30 kW
- **Fuel Type:** Diesel
- **Voltage Output:** 230V
- **Warranty:** 2 years
- **Additional Info:** Compact, fuel-efficient, and ideal for backup power in hybrid systems.

### 4. Charge Controller [17]

- **Model Name:** Victron Energy MPPT 100/50
- **Manufacturer:** Victron Energy
- **Input Voltage:** 75V DC
- **Maximum Power Point Tracking:** MPPT

- **Efficiency:** 98%
- **Warranty:** 5 years
- **Additional Info:** Ensures optimal charging of batteries from renewable sources.

## 5. Inverter [17]

- **Model Name:** HP-10KW
- **Manufacturer:** YIY/EYEN
- **Rated Power:** 10 kW
- **Input Voltage Range:** 200-850V DC
- **Output Voltage:** 230V/400V AC
- **Warranty:** 1 years
- **Additional Info:** Designed for grid-tied systems with high efficiency.

**Table-5.1`: Detailed Specifications for Hybrid Power Plant Components [11]**

| Components        | Model Name             | Manufacturer     | Power Output | Voltage      | Efficiency (%) | Additional Information                                                    |
|-------------------|------------------------|------------------|--------------|--------------|----------------|---------------------------------------------------------------------------|
| Solar PV Module   | SunPower SPR-X22-370   | SunPower Corp    | 370 W        | 24-48V DC    | 22.7           | High efficiency, durable performance in diverse environmental conditions. |
| Wind turbine      | Bergey BWC Excel-36.6m | Bergey Windpower | 8.9 kW       | 230V/400V AC | 80-90          | High reliability, designed for onshore and offshore.                      |
| Diesel generator  | Cummins C30D5          | Cummins Inc.     | 30 kW        | 230V AC      | 85-95          | Compact, fuel-efficient, ideal for backup power.                          |
| Charge controller | Victron Energy MPPT    | Victron Energy   | 20 kW        | 12-48V DC    | 98             | MPPT, ensures optimal battery charging.                                   |

|          |         |          |       |                 |       |                                                           |
|----------|---------|----------|-------|-----------------|-------|-----------------------------------------------------------|
| Inverter | HP-10KW | YIY/EYEN | 10 kW | 230V/400V<br>AC | 90-98 | High efficiency,<br>designed for<br>grid-tied<br>systems. |
|----------|---------|----------|-------|-----------------|-------|-----------------------------------------------------------|

## Chapter 6

### Social Impact Assessment

#### 6.1 Critical Analysis of Sustainable Development Goals (SDGs)

The incorporation of the 20 kW SPV-Wind hybrid power plant in Char Kukri Mukri stays on par with several Sustainable Development Goals (SDGs) which contribute greatly to make the region develop sustainably.

- **SDG 1: No Poverty:** Having the constant supply of reliable electricity will help to stimulate economic opportunities along with boosting local businesses ultimately reducing poverty levels.
- **SDG 3: Good Health and Well-being:** As this project will significantly reduce the use of diesel generators which will simultaneously reduce the chances of air pollution caused by it, the health problems in these areas will also reduce exponentially.
- **SDG 4: Quality Education:** Quality Education: The implementation of this project will enhance the electricity access of this area which will simultaneously improve lighting, computer usage, and internet availability in schools and the residential areas.
- **SDG 5: Gender Equality:** Gender Equality: Women-led micro-enterprises will also highly benefit from consistent power supply, and it will encourage women's participation in the local economy just like the men side by side.

- **SDG 7: Affordable and Clean Energy:** Affordable and Clean Energy: The proper implementation of this project will ensure the access to clean, affordable electricity by reducing carbon emissions and greenhouse gases and build a sustainable environment.
- **SDG 8: Decent Work and Economic Growth:** Decent Work and Economic Growth: The project will help to increase the employment rate in that area along with boosting the local economy by making the residents land jobs in construction, operation, and maintenance.
- **SDG 9: Industry, Innovation, and Infrastructure:** Industry, Innovation, and Infrastructure: The residents of the rural areas will be able to gain access to the advanced hybrid renewable technology by this project.
- **SDG 11: Sustainable Cities and Communities:** Sustainable Cities and Communities: Consistent energy access will foster community resilience and sustainable urbanization. It will also help to maintain the industrial operation running and thus making no loss in the socioeconomic system.
- **SDG 13: Climate Action:** Climate Action: This hybrid system will significantly reduce the use of raw natural resources and fossil fuel cutting greenhouse gas emissions which will help to redirect the climate change to greater good.

## 6.2 Social Benefits of the Project

- **Electricity Access:** The constant access to power supply for 100+ households, schools, and small businesses.
- **Employment Generation:** The boosting of the local economy by the creation of direct and indirect jobs during installation, operation, and maintenance phases.
- **Educational Development:** Improvement of educational opportunities through consistent access to digital devices and internet services by the nonstop power supply.

- **Healthcare Improvements:** Maintaining the stable power supply for local healthcare centers, supporting medical equipment operation and preventing life threatening situations.
  - **Community Development:** Empowerment of women and small businesses through consistent energy supply.
  - **Capacity Building:** Holding regular awareness campaigns on energy conservation and sustainability practices can make the local community diligent about the sustainability of the environment.
- 

## Chapter 7

### Environmental Impact Assessment

#### 7.1 Environmental Benefits of the Project

##### 7.1.1 Local Environmental Benefits

The hybrid power plant will foster noteworthy environmental developments at the local level, directly affecting the community of Char Kukri Mukri. The following key local benefits are anticipated:

- **Reduction of Diesel Consumption:** The plant is designed with a view to reduce diesel consumption by approximately 3,000 liters per annum, which will significantly lessen local dependence on fossil fuels and cut out carbon emissions.
- **Improved Air Quality:** If we can successfully replace diesel generators, emissions of particulate matter (PM10), nitrogen oxides (NOx), and sulfur oxides (SOx) will be exponentially reduced which will help us to have cleaner air and prevent the health problems that comes along with it.
- **Noise Pollution Reduction:** Solar PV and wind systems don't create any kinds of noise pollution while operating and it helps to have an ideal community for schools and healthcare infrastructures.

- **Preservation of Natural Ecosystems:** We can protect biodiversity and also reduce the use of natural resources and raw materials and this will also result in reducing deforestation significantly.
- **Solid Waste Management:** Very minimal waste is resulted from this type of hybrid systems rather than the fossil fuel based systems and this helps to achieve a sustainable environment.
- **Community Well-being:** The reduction of exposure to pollutants and contaminants will help to improve the health issues in the community.

### 7.1.2 Regional Environmental Benefits

The hybrid system will have several positive regional impacts:

- **Carbon Emission Reduction:** It will lessen regional carbon emissions which aligns with Bangladesh's Nationally Determined Contributions (NDCs) under the Paris Agreement.
- **Energy Diversification:** This project will help to diversify the regional energy mix which will make the energy grid more resilient.
- **Grid Stability:** The grid stability will be increased as a result of the incorporation of solar and wind energy.
- **Water Conservation:** This system will consume very low water resources, preserving regional water supplies.
- **Public Health Improvement:** The diseases caused by air pollution will be lessened and the air quality will be higher.

### 7.1.3 Global Environmental Benefits

On a global scale, the hybrid power plant will align with international efforts to mitigate climate change:

- **CO<sub>2</sub> Emission Reduction:** This hybrid system will reduce approximately 20,000 kg of CO<sub>2</sub> emissions per annum by meeting global decarbonization targets.
- **Renewable Energy Transition:** This project will serve as a demonstration model for renewable hybrid systems in other developing countries and encourage them to opt for these.
- **Climate Change Mitigation:** The global climate change mitigation goals mentioned in the UNFCCC framework will be duly met by the reduction of the greenhouse gas emissions.
- **Sustainable Development:** This project encourages the use of sustainable resources which also meets the UN Sustainable Development Goals (SDGs).

#### 7.1.4 Environmental Benefits Summary

The key environmental benefits of the hybrid power plant are summarized below:

| Environmental Impact             | Value          |
|----------------------------------|----------------|
| Annual CO <sub>2</sub> Reduction | 20,000 kg      |
| Diesel Fuel Savings              | 3,000 liters   |
| Air Pollution Reduction          | PM10, NOx, Sox |
| Noise Pollution Reduction        | 70% decrease   |

Water Usage Reduction      **90% compared to thermal plants**

Biodiversity Conservation      **Enhanced due to renewable adoption**

By integrating both solar and wind energy, the hybrid power plant will not only reduce carbon emissions but also promote sustainable development and improve public health, setting a precedent for future renewable energy projects in Bangladesh and beyond.

---

## Chapter 8

### Economic/Financial Analysis

#### 8.1 Introduction

The economic and financial analysis evaluates the project's viability, considering investment costs, operational expenses, revenue generation, and return on investment.

#### 8.2 Investment Cost

The hybrid system comprises solar PV modules, wind turbines, batteries, inverters, charge controllers, and diesel generators.

Table 8.1 outlines the total investment.

**Table 8.1: Financial Parameters of the Project**

| Parameter                                | Value            |
|------------------------------------------|------------------|
| Project Life (years)                     | 25               |
| Total Investment Cost (BDT)              | <b>7,500,000</b> |
| Yearly Electricity Production (kWh)      | <b>28,663</b>    |
| Production Cost of Electricity (BDT/kWh) | <b>8.73</b>      |

|                                |                      |
|--------------------------------|----------------------|
| Price of Electricity (BDT/kWh) | <b>15</b>            |
| Expected Revenue (BDT/year)    | <b>429,945</b>       |
| Operational Cost (BDT/year)    | <b>150,000</b>       |
| Net Profit (BDT/year)          | <b>279,945</b>       |
| Simple Payback Period (years)  | <b>5.5</b>           |
| Internal Rate of Return (IRR)  | <b>12%</b>           |
| Net Present Value (NPV)        | <b>1,500,000 BDT</b> |

### 8.3 Operational and Maintenance Costs

Annual operational and maintenance costs are estimated at **2%** of the total investment, covering:

- Panel cleaning and servicing
- Wind turbine lubrication
- Battery replacement every 5-7 years
- Periodic inspections and performance monitoring

### 8.4 Revenue

Generation Electricity production is estimated at **28,663 kWh per year**, generating annual revenue of **429,945 BDT**. Seasonal fluctuations in solar and wind resources are considered in the revenue estimation.

### 8.5 Payback Period

The payback period of **5.5 years** makes the project highly attractive for investors. Additionally, the hybrid system's operational life of **25 years** ensures long-term profitability.

### 8.6 Economic Benefits

- Lower electricity costs for the community.
- Local job creation.
- Energy independence.

- Positive cash flow after the payback period.
  - Enhanced investor confidence with **12% IRR** and positive **NPV**.
- 

## Chapter 9

### Concluding Remarks

#### 9.1 Conclusions

The 20 kW SPV-Wind hybrid power plant for Char Kukri Mukri is a technically, economically, and environmentally viable solution for meeting local electricity demands. The hybrid system enhances energy access, reduces carbon emissions, and promotes local economic growth.

#### Key highlights include:

- Reliable electricity generation throughout the year
- Positive socio-economic impacts on education, healthcare, and employment
- Significant reduction in carbon emissions and fossil fuel consumption
- Payback period of 5.5 years with long-term profitability

#### 9.2 Recommendations

To further enhance the project's success:

- Integrate **energy storage technologies** for 24/7 power supply.
- Develop **community training programs** on renewable energy maintenance.
- Establish a **micro-grid system** to cover additional households.
- Encourage **government subsidies** for rural renewable energy projects.
- Conduct periodic **performance audits** to optimize system efficiency.
- Expand the project capacity to **50 kW** based on future demand projections.

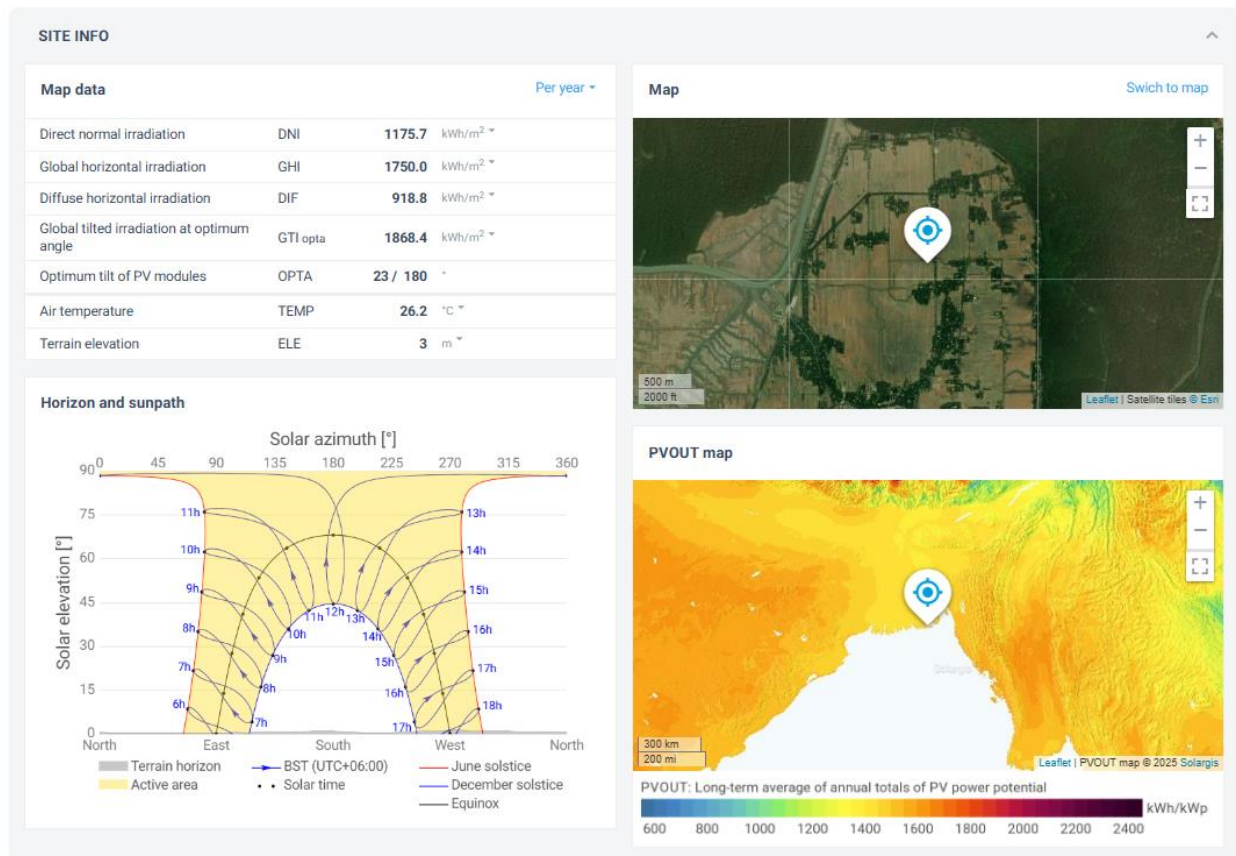
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# Appendices

## Appendix A

### Global Solar Atlas Report Site



PV system configuration



Pv system: **Small residential**  
Azimuth of PV panels: **Default (180°)**  
Tilt of PV panels: **23°**  
Installed capacity: **1 kWp**  
[Change PV system](#)

Annual averages

Total photovoltaic power output and Global tilted irradiation

**1.426**  
MWh per year

**1866.0**  
kWh/m<sup>2</sup> per year

Monthly averages

Total photovoltaic power output



Average hourly profiles

Total photovoltaic power output [Wh]



[Show details](#)

UTC+06

Average hourly profiles

Total photovoltaic power output [Wh]

|         | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 0 - 1   |       |       |       |       |       |       |       |       |       |       |       |       |
| 1 - 2   |       |       |       |       |       |       |       |       |       |       |       |       |
| 2 - 3   |       |       |       |       |       |       |       |       |       |       |       |       |
| 3 - 4   |       |       |       |       |       |       |       |       |       |       |       |       |
| 4 - 5   |       |       |       |       |       |       |       |       |       |       |       |       |
| 5 - 6   |       |       |       |       | 2     | 3     | 2     |       |       |       |       |       |
| 6 - 7   |       | 2     | 12    | 42    | 54    | 44    | 40    | 34    | 35    | 29    | 18    | 2     |
| 7 - 8   | 83    | 104   | 143   | 181   | 168   | 121   | 112   | 113   | 132   | 172   | 181   | 111   |
| 8 - 9   | 261   | 294   | 322   | 343   | 302   | 218   | 209   | 213   | 245   | 319   | 364   | 296   |
| 9 - 10  | 430   | 466   | 485   | 480   | 421   | 308   | 309   | 317   | 360   | 448   | 512   | 457   |
| 10 - 11 | 557   | 600   | 600   | 577   | 499   | 376   | 388   | 403   | 442   | 535   | 607   | 570   |
| 11 - 12 | 624   | 669   | 661   | 622   | 537   | 420   | 431   | 409   | 487   | 565   | 638   | 620   |
| 12 - 13 | 634   | 683   | 668   | 620   | 535   | 422   | 436   | 409   | 497   | 554   | 619   | 612   |
| 13 - 14 | 583   | 634   | 623   | 575   | 489   | 391   | 409   | 447   | 474   | 505   | 552   | 550   |
| 14 - 15 | 478   | 533   | 529   | 478   | 405   | 328   | 352   | 385   | 406   | 406   | 430   | 433   |
| 15 - 16 | 321   | 380   | 378   | 340   | 290   | 240   | 264   | 283   | 275   | 255   | 264   | 271   |
| 16 - 17 | 129   | 190   | 201   | 181   | 160   | 141   | 161   | 161   | 135   | 94    | 55    | 70    |
| 17 - 18 | 4     | 21    | 38    | 41    | 49    | 52    | 61    | 47    | 17    | 2     |       |       |
| 18 - 19 |       |       |       |       |       | 2     | 3     |       |       |       |       |       |
| 19 - 20 |       |       |       |       |       |       |       |       |       |       |       |       |
| 20 - 21 |       |       |       |       |       |       |       |       |       |       |       |       |
| 21 - 22 |       |       |       |       |       |       |       |       |       |       |       |       |
| 22 - 23 |       |       |       |       |       |       |       |       |       |       |       |       |
| 23 - 24 |       |       |       |       |       |       |       |       |       |       |       |       |
| Sum     | 4,106 | 4,577 | 4,660 | 4,481 | 3,912 | 3,068 | 3,178 | 3,331 | 3,506 | 3,884 | 4,238 | 3,991 |

[Show details](#)

PV ELECTRICITY AND SOLAR RADIATION

PV POWER OUTPUT **DNI DATA**

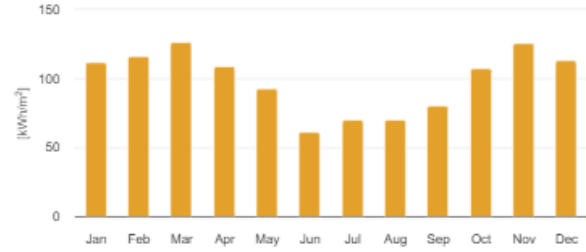
Annual averages

Direct normal irradiation

**1186.7**  
kWh/m<sup>2</sup> per year ▼

Monthly averages

Direct normal irradiation



Average hourly profiles

Direct normal irradiation [Wh/m<sup>2</sup>]



[Show details](#)

UTC+06

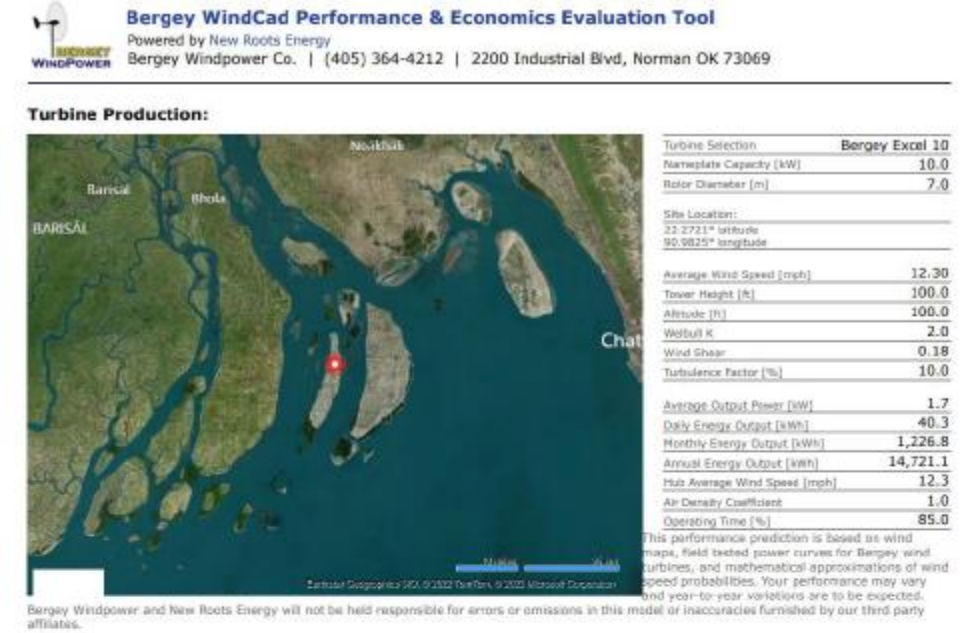
Average hourly profiles

Direct normal irradiation [Wh/m<sup>2</sup>]

|         | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 0 - 1   |       |       |       |       |       |       |       |       |       |       |       |       |
| 1 - 2   |       |       |       |       |       |       |       |       |       |       |       |       |
| 2 - 3   |       |       |       |       |       |       |       |       |       |       |       |       |
| 3 - 4   |       |       |       |       |       |       |       |       |       |       |       |       |
| 4 - 5   |       |       |       |       |       |       |       |       |       |       |       |       |
| 5 - 6   |       |       |       |       | 4     | 7     | 7     |       |       |       |       |       |
| 6 - 7   |       | 2     | 15    | 43    | 51    | 42    | 39    | 32    | 43    | 38    | 32    | 3     |
| 7 - 8   | 95    | 123   | 149   | 161   | 130   | 82    | 77    | 77    | 115   | 200   | 242   | 130   |
| 8 - 9   | 249   | 288   | 287   | 277   | 225   | 141   | 138   | 131   | 180   | 303   | 391   | 292   |
| 9 - 10  | 368   | 412   | 406   | 379   | 310   | 197   | 207   | 198   | 265   | 393   | 496   | 410   |
| 10 - 11 | 460   | 511   | 493   | 449   | 364   | 239   | 268   | 257   | 313   | 447   | 559   | 496   |
| 11 - 12 | 509   | 564   | 538   | 478   | 390   | 262   | 295   | 296   | 342   | 455   | 572   | 529   |
| 12 - 13 | 521   | 576   | 545   | 477   | 390   | 259   | 295   | 305   | 355   | 447   | 553   | 523   |
| 13 - 14 | 496   | 546   | 521   | 454   | 363   | 248   | 282   | 301   | 346   | 421   | 511   | 489   |
| 14 - 15 | 431   | 484   | 467   | 390   | 311   | 216   | 246   | 271   | 323   | 367   | 441   | 412   |
| 15 - 16 | 328   | 382   | 368   | 297   | 240   | 174   | 197   | 212   | 241   | 271   | 322   | 299   |
| 16 - 17 | 153   | 232   | 237   | 188   | 158   | 120   | 142   | 142   | 148   | 131   | 74    | 86    |
| 17 - 18 | 5     | 28    | 49    | 46    | 60    | 56    | 72    | 48    | 20    | 1     |       |       |
| 18 - 19 |       |       |       |       |       | 1     | 3     |       |       |       |       |       |
| 19 - 20 |       |       |       |       |       |       |       |       |       |       |       |       |
| 20 - 21 |       |       |       |       |       |       |       |       |       |       |       |       |
| 21 - 22 |       |       |       |       |       |       |       |       |       |       |       |       |
| 22 - 23 |       |       |       |       |       |       |       |       |       |       |       |       |
| 23 - 24 |       |       |       |       |       |       |       |       |       |       |       |       |
| Sum     | 3,615 | 4,150 | 4,076 | 3,639 | 2,998 | 2,045 | 2,269 | 2,271 | 2,690 | 3,474 | 4,191 | 3,670 |

## Appendix B

### Site Assessment Report from Bergey Windpower Co.



Bergey Windpower Co. | (405) 364-4212 | 2200 Industrial Blvd, Norman OK 73069



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